

Understanding, Identifying, and Preventing Phishing Attacks

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Agenda

- Introduction to Phishing
- Types of Phishing Attacks
- Recognizing Phishing Emails
- Identifying Fake Websites
- Social Engineering Techniques
- Real-World Examples
- Best Practices for Prevention
- Interactive Quiz
- Summary

What is Phishing?

Definition

Phishing is a cyberattack where attackers pretend to be trusted organizations or individuals to trick people into revealing sensitive information.

Attackers Target

Username

Passwords

Banking Information

Credit Card Numbers

OTPs

Personal Data

Common Communication Channels

Email

SMS

Phone Calls

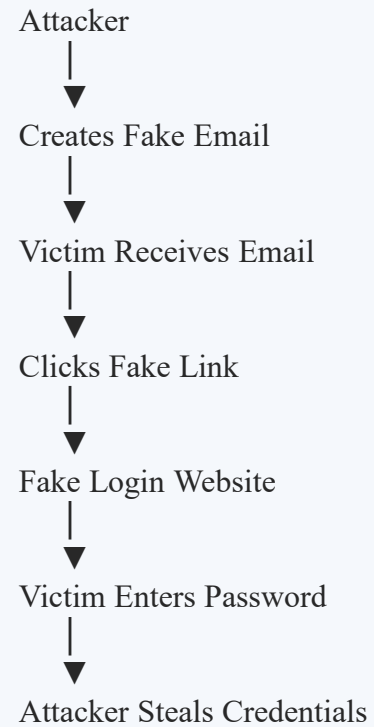
Social Media

Fake Websites

Example:

"Your bank account has been suspended. Click here to verify your account."

How Phishing Works



**Attackers rely on trust, urgency, and fear—
not technical hacking.**

Check for:

- Check HTTPS
- Verify Correct domain spelling
- Lock icon
- Professional design

Types of Phishing

Common Types of Phishing

1) Email Phishing

Fake emails pretending to be from banks, Amazon, Microsoft, etc.

2)Spear Phishing

Personalized attacks targeting specific individuals.

3)Whaling

Targets CEOs and senior executives.

4)Smishing

Phishing through SMS messages.

5)Vishing

Phishing through phone calls.

6)Social Media Phishing

Fake profiles and messages on Facebook, Instagram, LinkedIn, etc.

How Phishing Works

Phishing Attack Process

1. Attacker creates a fake email or website.
2. Victim receives the message.
3. Victim clicks the malicious link.
4. Victim enters login credentials.
5. Credentials are sent to the attacker.
6. Attacker gains unauthorized access.

Result: Data theft, financial loss, or identity theft

Warning Signs

- Unknown sender
- Urgent or threatening language
- Spelling and grammar mistakes
- Suspicious links
- Unexpected attachments
- Requests for passwords or OTPs
- Generic greetings like "Dear Customer"

Example

✗ Your account will be suspended. Click here immediately.

How to Identify Fake Websites

Check the website URL carefully.

Example:

✓ <https://www.amazon.com>

✗ <https://amazon-login-security.com>

Look for:

- HTTPS
- Correct spelling
- Company logo
- Contact information
- Valid certificates
- Never trust a website only because it has a lock icon.

Common Techniques Used by Attackers

1. Impersonation

Pretending to be your manager or IT support.

2. Pretexting

Creating a fake story to gain trust.

3. Baiting

Offering free software or USB drives.

4. Scareware

Showing fake virus warnings.

5. Quid Pro Quo

Offering help in exchange for confidential information.

Example 1: Bank Phishing

Email:

"Your SBI account has been blocked. Verify immediately."

Victim clicks the link.

Victim enters banking credentials.

Attacker steals money.

Example 2: Delivery Scam

SMS:

"Your parcel delivery failed. Pay ₹50 to reschedule."

Victim enters debit card details.

Money gets stolen.

Stay Safe from Phishing

Always:

- Verify the sender's email address.
- Hover over links before clicking.
- Use Multi-Factor Authentication (MFA).
- Use strong passwords.
- Update software regularly.
- Install antivirus software.
- Never share OTPs.
- Report suspicious emails immediately.

Immediate Actions

1. Disconnect from the internet.
2. Change your password.
3. Enable MFA.
4. Inform your IT department.
5. Scan your computer.
6. Contact your bank if financial information was shared.
7. Monitor your accounts.

Question 1

Which of the following is a phishing indicator?

- A. Correct company domain
- B. Urgent request for password
- C. Secure website
- D. Official customer support

Answer: B

Question 2

Should you click unknown email links?

- A. Yes
- B. No

Answer: No

Question 3

Which is safer?

- A. Same password everywhere
- B. Strong unique passwords

Answer: B

Question 4

True or False

HTTPS always means a website is safe.

Answer: False

Before clicking any link, ask yourself:

- ✓ Do I know the sender?
- ✓ Is the URL correct?
- ✓ Is the email asking for personal information?
- ✓ Is there urgency?
- ✓ Are there grammar mistakes?
- ✓ Does the message seem suspicious?

If yes, **don't click.**

Key Takeaways

- Think before you click.
- Verify every email.
- Check website URLs carefully.
- Never share passwords or OTPs.
- Use Multi-Factor Authentication.
- Report phishing attempts immediately.

Remember:

"Stop • Think • Verify • Click"

- ❖ National Institute of Standards and Technology (NIST)
- ❖ Cybersecurity and Infrastructure Security Agency (CISA)
- ❖ Microsoft Security
- ❖ Google Safety Center
- ❖ OWASP Phishing Awareness Resources

Thank You

Stay Safe Online.